Power input (USB−C) and regulators

File: lcd_dual_power.kicad_sch

HDMI receiver LT8619C

File: lcd_dual_hdmi.kicad_sch

AT070TN92 panel, bias supplies, VCOM, backlight

File: lcd_dual_panel.kicad_sch

STM32F072 controller and GT915 touch

File: lcd_dual_mcu.kicad_sch

AT070TN92 7'' 800x480 TTL panel driver: HDMI (LT8619C) + GT915 touch −> USB HID (STM32F072).

Power: USB−C 5 V (from Raspberry Pi USB port, ~0.7 A max with backlight at 100 %).

U10: STM32F072CBT6 or CH32X035C8T6 (same LQFP−48 footprint). SW2 = CH32 USB bootloader (D+ high at power−up).

Panel power−up order (MCU controlled): DVDD −> RESET −> VGL/AVDD −> VGH −> data −> backlight.

J3 pad numbers equal AT070TN92 FPC pin numbers; use a TOP−contact 0.5 mm 50−pin FPC connector.

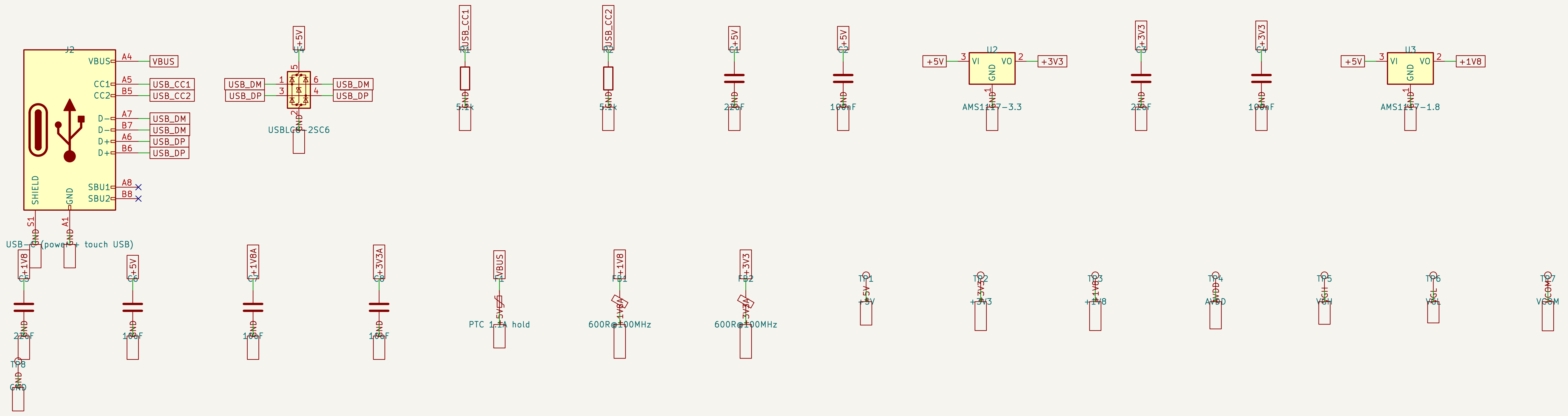
VCOM (pins 6/46) is adjustable with RV1 (2.55..4.5 V): trim for minimum flicker.

LT8619C HDMI−>RGB888, TPS61040 bias, TPS61165 backlight, STM32F072 USB touch

Sheet: /
File: lcd_dual.kicad_sch

Title: AT070TN92 HDMI + USB−touch driver board

Size: A4
Date: 2026−10−02
Rev: 1.0
KiCad E.D.A. kicad−cli 7.0.11+dfsg−1build4
Id: 1/5


J2
VBUS
A4
VBUS
CC1
A5
USB_CC1
CC2
B5
USB_CC2
D−
A7
USB_DM
D−
B7
USB_DM
D+
A6
USB_DP
D+
B6
USB_DP
SBU1
A8
SBU2
B8
SHIELD
GND
S1
A1
GND
GND
USB-C (power + touch USB)
+5V
U4
5
USB_DM
1
USB_DP
3
6
USB_DM
4
USB_DP
2
GND
USBLC6-2SC6
USB_CC1
R1
GND
5.1k
USB_CC2
R2
GND
5.1k
+5V
C1
GND
22uF
+5V
C2
GND
100nF
+5V
3
VI
U2
VO
2
+3V3
GND
1
GND
AMS1117-3.3
+3V3
C3
GND
22uF
+3V3
C4
GND
100nF
+5V
3
VI
U3
VO
2
+1V8
GND
1
GND
AMS1117-1.8
+1V8
C5
GND
22uF
TP8
GND
GND
+5V
C6
GND
10uF
+1V8A
C7
GND
10uF
+3V3A
C8
GND
10uF
VBUS
F1
+5V
PTC 1.1A hold
+1V8
FB1
+1V8A
600R@100MHz
+3V3
FB2
+3V3A
600R@100MHz
TP1
+5V
+5V
TP2
+3V3
+3V3
TP3
+1V8
+1V8
TP4
AVDD
AVDD
TP5
VGH
VGH
TP6
VGL
VGL
TP7
VCOM
VCOM

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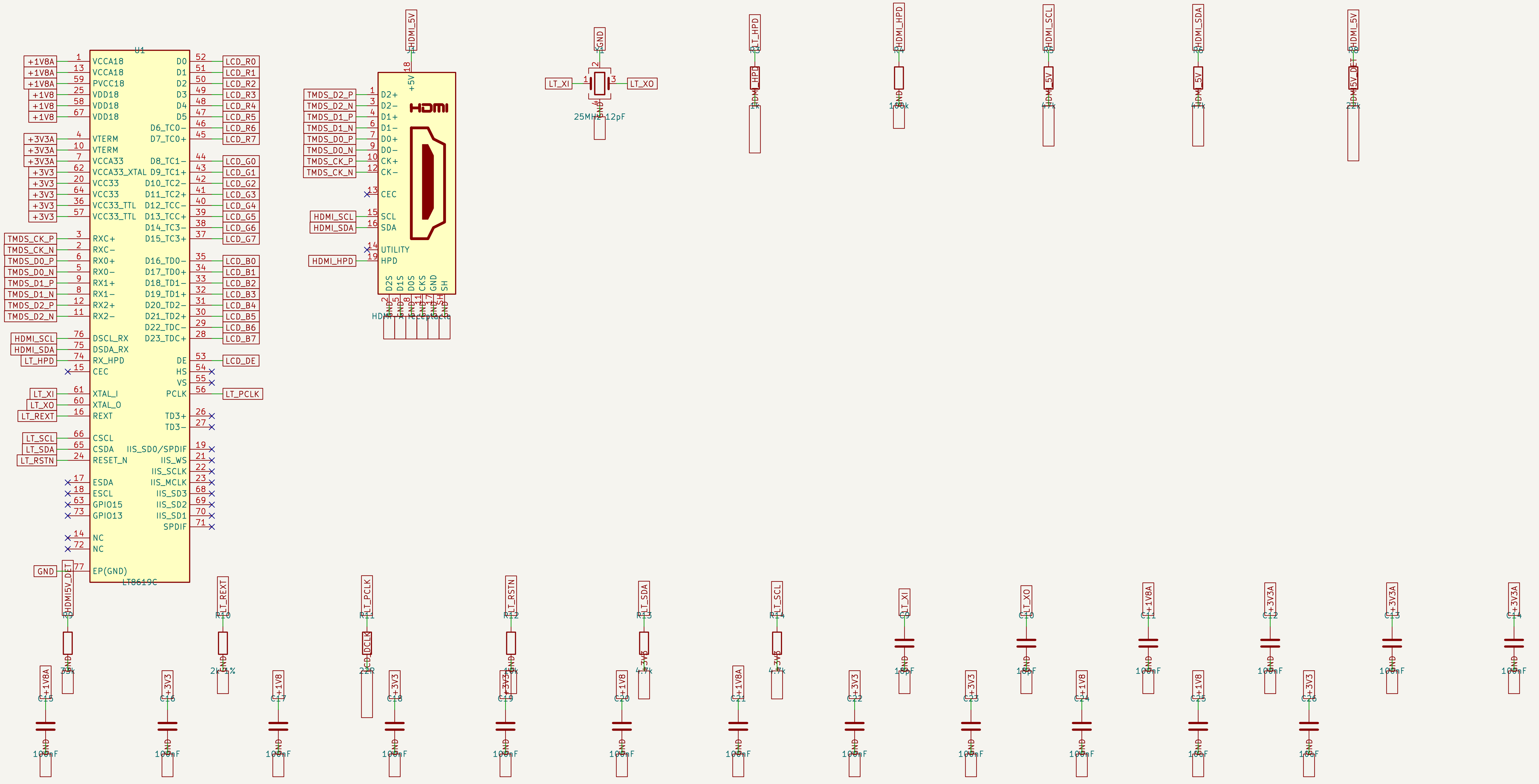
AT070TN92 HDMI + USB touch driver

Sheet: /Power input (USB−C) and regulators/
File: lcd_dual_power.kicad_sch

Title: Power input (USB−C) and regulators

Size: A3 Date: 2026-10-02 **Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 Id: 2/5


U1
LT8619C
+1V8A 1 VCCA18
+1V8A 13 VCCA18
+1V8A 59 PVCC18
+1V8 25 VDD18
+1V8 58 VDD18
+1V8 67 VDD18
+3V3A 4 VTERM
+3V3A 10 VTERM
+3V3A 7 VCCA33
+3V3 62 VCCA33_XTAL
+3V3 20 VCC33
+3V3 64 VCC33
+3V3 36 VCC33_TTL
+3V3 57 VCC33_TTL
TMDS_CK_P 3 RXC+
TMDS_CK_N 2 RXC−
TMDS_D0_P 6 RX0+
TMDS_D0_N 5 RX0−
TMDS_D1_P 9 RX1+
TMDS_D1_N 8 RX1−
TMDS_D2_P 12 RX2+
TMDS_D2_N 11 RX2−
HDMI_SCL 76 DSCL_RX
HDMI_SDA 75 DSDA_RX
LT_HPD 74 RX_HPD
15 CEC
LT_XI 61 XTAL_I
LT_XO 60 XTAL_O
LT_REXT 16 REXT
LT_SCL 66 CSCL
LT_SDA 65 CSDA
LT_RSTN 24 RESET_N
17 ESDA
18 ESCL
63 GPIO15
73 GPIO13
14 NC
72 NC
GND 77 EP(GND)
D0 52 LCD_R0
D1 51 LCD_R1
D2 50 LCD_R2
D3 49 LCD_R3
D4 48 LCD_R4
D5 47 LCD_R5
D6_TC0− 46 LCD_R6
D7_TC0+ 45 LCD_R7
D8_TC1− 44 LCD_G0
D9_TC1+ 43 LCD_G1
D10_TC2− 42 LCD_G2
D11_TC2+ 41 LCD_G3
D12_TCC− 40 LCD_G4
D13_TCC+ 39 LCD_G5
D14_TC3− 38 LCD_G6
D15_TC3+ 37 LCD_G7
D16_TD0− 35 LCD_B0
D17_TD0+ 34 LCD_B1
D18_TD1− 33 LCD_B2
D19_TD1+ 32 LCD_B3
D20_TD2− 31 LCD_B4
D21_TD2+ 30 LCD_B5
D22_TDC− 29 LCD_B6
D23_TDC+ 28 LCD_B7
DE 53 LCD_DE
HS 54
VS 55
PCLK 56 LT_PCLK
TD3+ 26
TD3− 27
IIS_SD0/SPDIF 19
IIS_WS 21
IIS_SCLK 22
IIS_MCLK 23
IIS_SD3 68
IIS_SD2 69
IIS_SD1 70
SPDIF 71
J1
HDMI-A receptacle
HDMI_5V 18 +5V
TMDS_D2_P 1 D2+
TMDS_D2_N 3 D2−
TMDS_D1_P 4 D1+
TMDS_D1_N 6 D1−
TMDS_D0_P 7 D0+
TMDS_D0_N 9 D0−
TMDS_CK_P 10 CK+
TMDS_CK_N 12 CK−
13 CEC
HDMI_SCL 15 SCL
HDMI_SDA 16 SDA
14 UTILITY
HDMI_HPD 19 HPD
D2S 2 GND
D1S 5 GND
D0S 8 GND
CKS 11 GND
GND 17 GND
SH SH GND
HDMI
Y1
25MHz 12pF
LT_XI 1
3 LT_XO
2 GND
4 GND
R3 LT_HPD HDMI_HPD 1k
R4 HDMI_HPD GND 100k
R5 HDMI_SCL HDMI_5V 47k
R6 HDMI_SDA HDMI_5V 47k
R8 HDMI_5V HDMI5V_DET 22k
R9 HDMI5V_DET GND 33k
R10 LT_REXT GND 2k 1%
R11 LT_PCLK LCD_DCLK 22R
R12 LT_RSTN +3V3 10k
R13 LT_SDA +3V3 4.7k
R14 LT_SCL +3V3 4.7k
C9 LT_XI GND 18pF
C10 LT_XO GND 18pF
C11 +1V8A GND 100nF
C12 +3V3A GND 100nF
C13 +3V3A GND 100nF
C14 +3V3A GND 100nF
C15 +1V8A GND 100nF
C16 +3V3 GND 100nF
C17 +1V8 GND 100nF
C18 +3V3 GND 100nF
C19 +3V3 GND 100nF
C20 +1V8 GND 100nF
C21 +1V8A GND 100nF
C22 +3V3 GND 100nF
C23 +3V3 GND 100nF
C24 +1V8 GND 100nF
C25 +1V8 GND 10uF
C26 +3V3 GND 10uF

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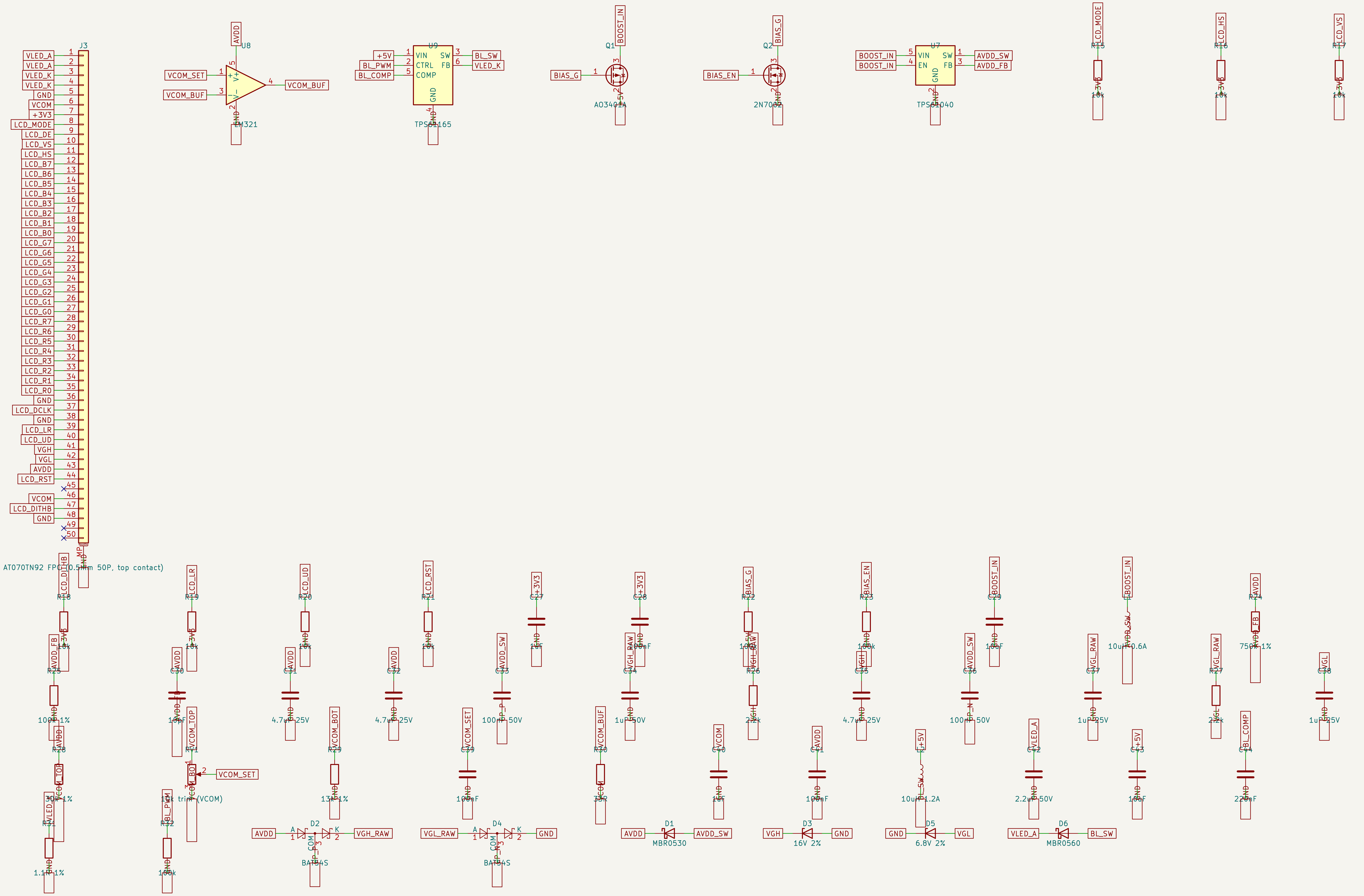
AT070TN92 HDMI + USB touch driver

Sheet: /HDMI receiver LT8619C/
File: lcd_dual_hdmi.kicad_sch

Title: HDMI receiver LT8619C

Size: A3
Date: 2026-10-02
Rev: 1.0
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 3/5


J3
VLED_A
VLED_A
VLED_K
VLED_K
GND
VCOM
+3V3
LCD_MODE
LCD_DE
LCD_VS
LCD_HS
LCD_B7
LCD_B6
LCD_B5
LCD_B4
LCD_B3
LCD_B2
LCD_B1
LCD_B0
LCD_G7
LCD_G6
LCD_G5
LCD_G4
LCD_G3
LCD_G2
LCD_G1
LCD_G0
LCD_R7
LCD_R6
LCD_R5
LCD_R4
LCD_R3
LCD_R2
LCD_R1
LCD_R0
GND
LCD_DCLK
GND
LCD_LR
LCD_UD
VGH
VGL
AVDD
LCD_RST
VCOM
LCD_DITHB
GND
AT070TN92 FPC (0.5mm 50P, top contact)
U8
AVDD
VCOM_SET
VCOM_BUF
VCOM_BUF
GND
LM321
U9
+5V
BL_PWM
BL_COMP
VIN
CTRL
COMP
SW
FB
BL_SW
VLED_K
GND
TPS61165
Q1
BOOST_IN
BIAS_G
+5V
AO3401A
Q2
BIAS_G
BIAS_EN
GND
2N7002
U7
BOOST_IN
BOOST_IN
VIN
EN
SW
FB
AVDD_SW
AVDD_FB
GND
TPS61040
R15
LCD_MODE
+3V3
10k
R16
LCD_HS
+3V3
10k
R17
LCD_VS
+3V3
10k
R18
LCD_DITHB
+3V3
10k
R19
LCD_LR
+3V3
10k
R20
LCD_UD
GND
10k
R21
LCD_RST
GND
10k
C27
+3V3
GND
1uF
C28
+3V3
GND
100nF
R22
BIAS_G
+5V
100k
R23
BIAS_EN
GND
100k
C29
BOOST_IN
GND
100nF
L1
BOOST_IN
AVDD_SW
10uH 0.6A
R24
AVDD
AVDD_FB
750k 1%
R25
AVDD_FB
GND
100k 1%
C30
AVDD
AVDD_FB
10pF
C31
AVDD
GND
4.7uF 25V
C32
AVDD
GND
4.7uF 25V
C33
AVDD_SW
CP_P
100nF 50V
C34
VGH_RAW
GND
1uF 50V
R26
VGH_RAW
VGH
22k
C35
VGH
GND
4.7uF 25V
C36
AVDD_SW
CP_N
100nF 50V
C37
VGL_RAW
GND
1uF 25V
R27
VGL_RAW
VGL
22k
C38
VGL
GND
1uF 25V
R28
AVDD
VCOM_TOP
30k 1%
RV1
VCOM_TOP
VCOM_BOT
VCOM_SET
10k trim (VCOM)
R29
VCOM_BOT
GND
13k 1%
C39
VCOM_SET
GND
100nF
R30
VCOM_BUF
VCOM
33R
C40
VCOM
GND
1uF
C41
AVDD
GND
100nF
L2
+5V
BL_SW
10uH 1.2A
C42
VLED_A
GND
2.2uF 50V
C43
+5V
GND
10uF
C44
BL_COMP
GND
220nF
R31
VLED_K
GND
1.1R 1%
R32
BL_PWM
GND
100k
D2
AVDD
VGH_RAW
CP_P
BAT54S
D4
VGL_RAW
GND
CP_N
BAT54S
D1
AVDD
AVDD_SW
MBR0530
D3
VGH
GND
16V 2%
D5
GND
VGL
6.8V 2%
D6
VLED_A
BL_SW
MBR0560

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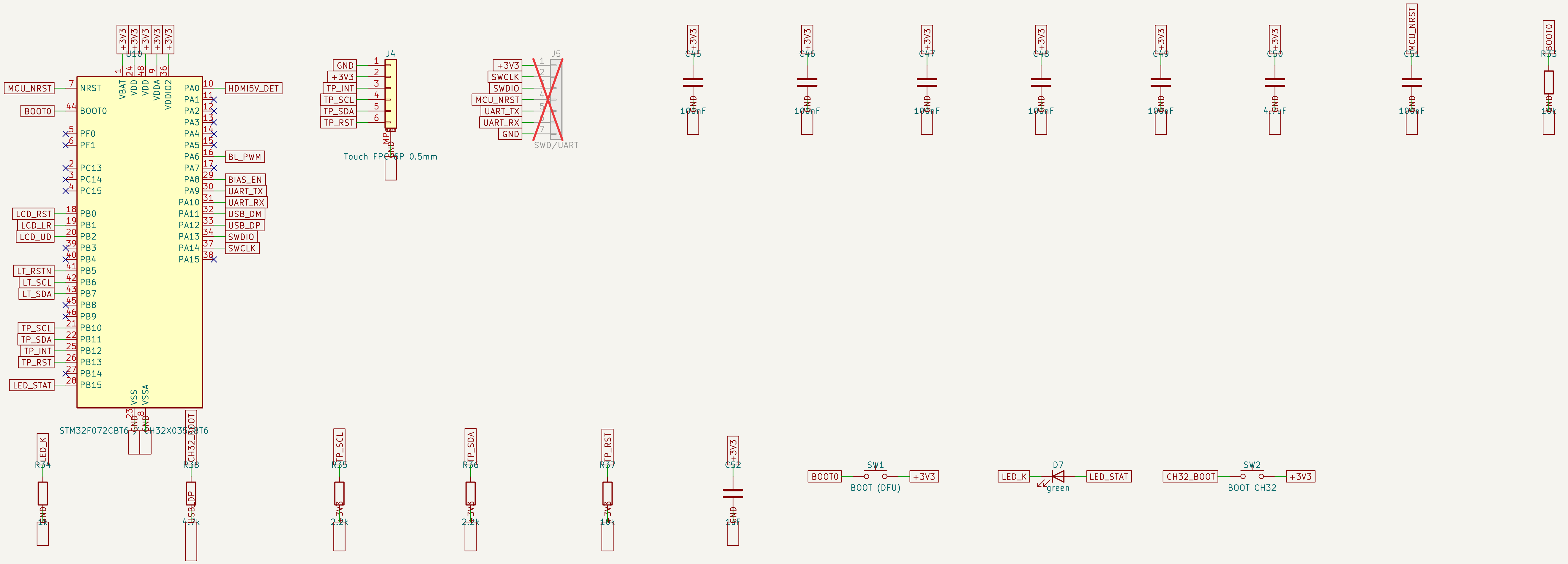
AT070TN92 HDMI + USB touch driver

Sheet: /AT070TN92 panel, bias supplies, VCOM, backlight/

File: lcd_dual_panel.kicad_sch

Title: AT070TN92 panel, bias supplies, VCOM, backlight

Size: A3	Date: 2026-10-02	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/5


U10
STM32F072CBT6 / CH32X035C8T6
+3V3
VBAT 1
VDD 24
VDD 48
VDDA 9
VDDIO2 36
MCU_NRST 7 NRST
BOOT0 44 BOOT0
5 PF0
6 PF1
2 PC13
3 PC14
4 PC15
LCD_RST 18 PB0
LCD_LR 19 PB1
LCD_UD 20 PB2
39 PB3
40 PB4
LT_RSTN 41 PB5
LT_SCL 42 PB6
LT_SDA 43 PB7
45 PB8
46 PB9
TP_SCL 21 PB10
TP_SDA 22 PB11
TP_INT 25 PB12
TP_RST 26 PB13
27 PB14
LED_STAT 28 PB15
PA0 10 HDMI5V_DET
PA1 11
PA2 12
PA3 13
PA4 14
PA5 15
PA6 16 BL_PWM
PA7 17
PA8 29 BIAS_EN
PA9 30 UART_TX
PA10 31 UART_RX
PA11 32 USB_DM
PA12 33 USB_DP
PA13 34 SWDIO
PA14 37 SWCLK
PA15 38
VSS 23 GND
VSSA 8 GND
J4
GND 1
+3V3 2
TP_INT 3
TP_SCL 4
TP_SDA 5
TP_RST 6
MP GND
Touch FPC 6P 0.5mm
J5
+3V3 1
SWCLK 2
SWDIO 3
MCU_NRST 4
UART_TX 5
UART_RX 6
GND 7
SWD/UART
C45 +3V3 GND 100nF
C46 +3V3 GND 100nF
C47 +3V3 GND 100nF
C48 +3V3 GND 100nF
C49 +3V3 GND 100nF
C50 +3V3 GND 4.7uF
C51 MCU_NRST GND 100nF
R33 BOOT0 GND 10k
R34 LED_K GND 1k
R38 CH32_BOOT USB_DP 4.7k
R35 TP_SCL +3V3 2.2k
R36 TP_SDA +3V3 2.2k
R37 TP_RST +3V3 10k
C52 +3V3 GND 1uF
SW1
BOOT0
+3V3
BOOT (DFU)
D7
LED_K
LED_STAT
green
SW2
CH32_BOOT
+3V3
BOOT CH32

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AT070TN92 HDMI + USB touch driver
Sheet: /STM32F072 controller and GT915 touch/
File: lcd_dual_mcu.kicad_sch

Title: STM32F072 controller and GT915 touch

Size: A3	Date: 2026-10-02	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/5